

MIE 262

Warp Shoe Project Report

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1. Abstract

This project report conducted its analysis into a series of shoe production process, combining theoretical models and tools obtained from course materials with our own assumptions and thinkings, aims to maximize the net profit of WARP Shoe Company.

2. Introduction

In this project we are asked to help WARP Shoe Company to gain maximum profit by constructing a LP model. At the beginning of 2006, a strong competitor of WARP went bankrupt. The market analyst in the WARP company predicted that demand for shoes on the market would double. The company aims to utilize this period by updating a profitable production plan. We will apply models and tools obtained in this operation research course to model the problem.

The whole problem is structured by a relaxed LP model. We will start by constructing the objective function and constraints to structure the model. We will be using AMPL to test and optimize feasible solutions for this model. Each variable value will be rounded up (if decimal part > 0.5) or rounded down otherwise to its closest integer, finally composing our production plan for this company. We will also give a analysis of strength and limitation of this model as an evaluation.

3. Methodology

3.1 Assumptions

Assumptions made in problem statement:

- closing inventory of January 2006 was zero for all types of shoes
- all sales happen at the end of the month
- the budget for raw materials is \$10,000,000
- setup times and costs are negligible
- workers are paid at \$25/hour
- each machine is to be operated by one worker
- transportation costs are negligible
- manufacturing sequence is negligible

Assumptions we made:

- for convenience, all units are measured regarding to seconds
- for convenience, the demand is averaged in computation
- warehouse rental fee is negligible
- individual capacity in warehouses is negligible

3.2 LP model (relaxed)

3.2.1 Sets

Purpose: We want to define the sets to be used in this model as below for each instance to iterate through:

- set **J**, the total types of shoes 557
- set **N**, the number of raw material
- set **E**, number of machine
- set **W**, the number of warehouse

3.2.2 Parameters (all lowercased letters are instances in corresponding sets listed in the above section)

Purpose: We want to set up parameters in this section so that each corresponding instance in the database can be extracted.

- **quantity{J,N}** : raw material **n** needed for **j** type of shoes, default at 0
- **cost{N}** : cost of each raw material **n**
- **avgDuration{J, E}** : the average time for each **j** type of shoe being machined at **e**, default at 0
- **opcost{E}** : the operating cost at **e**
- **sales{J}** : sales of **j** type of shoes
- **s_Quantity{N}** : number of **n** type of raw material
- **demand{J}** : demand for **j** type of shoes
- net capacity of warehouses **C**: 140,000
- net Machine operating time **T**: 20,160 minutes

3.2.3 Decision variables

$$X_{\{j\}} (\geq 0)$$

3.2.4 Objective function

By studying the project requirements we set up the objective function as in this relationship:

$$z = \text{Profit in sales} - \text{Raw material cost} - \text{Machine operating cost} - \text{Labor cost} - \text{Not meeting demand lost}$$

By defining sets, parameters and decision variables above, we are able to formulate individual sections as:

$$\text{Profit in sales} = \sum_{j=1}^{557} \text{sales}\{j\} \times X_{\{j\}}$$

$$\text{Raw material cost} = \sum_{j=1}^{557} \sum_{n=1}^7 X_{\{j\}} \times \text{quantity}\{j, n\} \times \text{cost}\{n\}$$

$$\text{Machine operating cost} = \sum_{j=1}^{557} \sum_{e=1}^8 \frac{1}{60} X_{\{j\}} \times \text{opcost}\{e\} \times \text{avgDuration}\{j, e\}$$

$$\text{Labor cost} = \sum_{j=1}^{557} \sum_{e=1}^8 \frac{25}{3600} X_{\{j\}} \times \text{avgDuration}\{j, e\}$$

$$\text{Not meeting demand lost} = \sum_{j=1}^{557} 10 (2 \times \text{demand}\{j\} - X_{\{j\}})$$

Finalizing our objective function we have:

$$\begin{aligned} z (\text{maximize}) = & \sum_{j=1}^{557} \text{sales}\{j\} \times X_{\{j\}} - \sum_{j=1}^{557} \sum_{n=1}^7 X_{\{j\}} \times \text{quantity}\{j, n\} \times \\ & \text{cost}\{n\} - \sum_{j=1}^{557} \sum_{e=1}^8 \frac{1}{60} X_{\{j\}} \times \text{opcost}\{e\} \times \text{avgDuration}\{j, e\} - \\ & \sum_{j=1}^{557} \sum_{e=1}^8 \frac{25}{3600} X_{\{j\}} \times \text{avgDuration}\{j, e\} - \sum_{j=1}^{557} 10 (2 \times \text{demand}\{j\} - \\ & X_{\{j\}}) \end{aligned}$$

3.2.5 Constraints

The above model needs to be tightened using the following constraints we identified from the problem statement:

- **Material :** $\sum_{j=1}^{557} \sum_{n=1}^7 X_{\{j\}} \times \text{quantity}\{j, n\} \times \text{cost}\{n\} \leq 10,000,000$
- **Capacity:** $\sum_{j=1}^{557} X_{\{j\}} \leq C(\text{net capacity of warehouses})$
- **Raw material** = $\sum_{j=1}^{557} X_{\{j\}} \times \text{quantity}\{j, n\} \leq s_Quantity\{n\}, \forall n \text{ in } N$
- **Machine production time:** $\sum_{j=1}^{557} \frac{1}{60} X_{\{j\}} \times \text{avgDuration}\{j, e\} \leq T, \forall e \text{ in } E$
- **Demand:** $\sum_{j=1}^{557} X_{\{j\}} \leq \text{demand}[j]$

4. Results

We utilized AMPL to optimize our model, with an optimal net profit for WARP Shoe Company at 11794732.99 dollars. (Production details of each type of shoes is in the attached .out file)

Clearly, since this is an ideal maximization problem, the result returned in reality should be expected to be lower than what we ended up using simulations with AMPL. The strength of this model is we clearly see the combination of the most profitable production plans, while it offers great modularity for the user to manipulate in future practices. However, one weakness being observed in this model is that in the first run we did not include the demand constraint for each type of shoes to be produced, where our model would only try to produce shoe types that are of higher unit profits, instead of taking individual demands into consideration. We recognized this error and introduced the demand constraint into the model to eliminate this “demand-production independency” error. Another weakness we have identified is this model actually simplifies the demand by simply taking an average, which might not be reliable in a larger scale computation as we are unable to match each product with precise demands, causing potential waste or losses.

5. Conclusion

In order to fulfill the profit needs for WRAP Shoe company in production, we utilized knowledge learned during this course by setting variables, parameters to access corresponding raw data and set up sub-models for different expenditures and revenues, thus composing a general optimization model for the entire problem. We used AMPL to solve the model and reached net profit at . We also discussed strength and weaknesses of our model where it could be improved in the future.

6. Answers to research questions

6.1 How should you estimate the demand for the month of February?

Answer : We took the data for demand by averaging them.

6.2 How many variables and constraints do you have?

Answer : There are 557 decision variables and 5 constraints.

6.3 If you had to relax your integer program to an LP, how many constraints were violated after rounding the LP solution to the closest integer solution?

Answer : 2 constraints are observed to be violated in demand and raw materials.

6.4 Which constraints are binding, and what is the real-world interpretation of those binding constraints?

Answer : Only demand and raw materials are binding.

6.5 Assume that some additional warehouse space is available at the price of \$10/box of shoes. Is it economical to buy it? What is the optimal amount of space to buy in this situation?

Answer : No. Since in the solution we found that warehouse capacity is not binding, meaning we would still have space for extra storage.

6.6 Imagine that machines were available for only 8 hours per day. How would

your solution change? Which constraints are binding now? Does the new solution seem realistic to you?

Answer : Now the operation hours are : $8 \times 28 \times 60 = 13440$ minutes in total, which returned us the same net profit as we get before, which is realistic.

6.7 If in addition there were a \$7,000,000 budget available to buy raw materials, what would you do? Change your formulation and solve again.

Answer : Since our raw materials is not binding at this optimal solution, it means we are not running out of them at the end of the production cycle, adding an extra 7,000,000 would not make changes to the existing solution.